INTRA-BASKET REVERSION

Abstract. We test the fifty round-5 products for within-basket statistical arbitrage using days 2 and 3 as formation data and day 4 as a locked out-of-sample sample. The evidence separates the universe into a structural identity, a compact constant multi-spread basket, several dynamic fair-value baskets, and three rejected baskets. PEBBLES is the only exact full-basket identity, but the full five-leg taker trade is dominated by bid-ask cost. SNACKPACK remains the strongest relationship space, while SLEEP POD, MICROCHIP, OXYGEN SHAKE, and TRANSLATOR contain usable dynamic residuals. The recommended implementation uses L1-normalized spread weights, formation-only z-score scaling, and a causal pre-update fair-value residual. Mid-price trading tests are reported to compare construction choices; transaction costs are ignored except where they overturn PEBBLES as a full-basket taker trade.

1. Protocol and statistical objects

There are ten baskets, each containing five products. For one basket, write the mid-price vector at time t as

$$p_t = (p_{1,t}, \dots, p_{5,t})^\top \in \mathbb{R}^5.$$

A candidate intra-basket spread is a linear synthetic price

$$s_t = w^\top p_t,$$

where w is fixed before day 4. Except for the PEBBLES identity, all displayed non-identity spreads are normalized by $\sum_i |w_i| = 1$. This normalization does not change z-scores, but it makes trading exposure and spread-unit PnL comparable across two-leg and four-leg trades.

The data protocol is deliberately strict. Days 2 and 3 are the formation sample F; day 4 is the locked out-of-sample sample O. Spread weights, deterministic trends, fair-value update gains, AR(1) parameters, residual scales, and thresholds are selected from F. The only operation allowed on O is sequential causal updating with a gain selected before O begins.

The first layer is relationship discovery. Let $\Delta p_t = p_t - p_{t-1}$ and let C_Δ be the correlation matrix of standardized formation price changes inside a basket. If the smallest principal-component share of C_Δ is close to zero, then a low-variance incremental relation exists. If all five principal-component shares are close to $1/5$, price changes are approximately five independent innovations and a dynamic filter alone is not sufficient evidence of arbitrage.

The second layer is fair-value modeling. The residual to trade is

$$u_t = s_t - m_t,$$

where m_t is the fair value of the spread. We compare three fair-value classes:

$$\begin{aligned} \text{static:} \quad & m_t = m, \\ \text{trend:} \quad & m_t = a + bt, \\ \text{causal dynamic:} \quad & m_{t+1|t} = m_{t|t-1} + K(s_t - m_{t|t-1}). \end{aligned}$$

For the dynamic model the traded residual is the pre-update innovation

$$u_t = s_t - m_{t|t-1},$$

not the smoothed residual after the update. This is the key anti-look-ahead convention.

Residuals are standardized with formation-only scale,

$$z_t = \frac{u_t - \mu_F}{\sigma_F}, \qquad \mu_F = \frac{1}{|F|}\sum_{t\in F} u_t, \qquad \sigma_F^2 = \frac{1}{|F|-1}\sum_{t\in F}(u_t - \mu_F)^2.$$

The diagnostic AR(1) model is

$$u_{t+1} = c + \phi u_t + \varepsilon_{t+1},$$

with half-life $h = -\log 2/\log \phi$ when $0 < \phi < 1$. We also report an augmented Dickey-Fuller p-value on the formation residual as a unit-root rejection test. The ADF test is not used alone for promotion, because a dynamic filter can mechanically make weak baskets look stationary. Promotion requires basket-level structure, out-of-sample z-score calibration, and a reasonable trading expression.

2. Discovery evidence

Table 1 gives the formation-sample basket diagnostics. SNACKPACK and PEBBLES are the only baskets with a very small price-change null direction. SNACKPACK has a smallest price-change PC share of 0.006, while PEBBLES has 0.002. The remaining baskets have smallest price-change PC shares near 0.195, essentially the independent five-dimensional benchmark 1/5. This proves that the strict constant/incremental test finds only SNACKPACK and PEBBLES as core compact relationships. The secondary baskets are therefore not promoted because of price-change PCA; they are promoted only if level relationships and dynamic residuals survive OOS.

Table 1. Formation-sample basket diagnostics. The price-change null direction is measured by the smallest principal-component share of standardized price changes. Values near 0.2 mean no compact incremental relation in a five-product basket.

Basket	Δ PC1	Δ min	Level PC1	Level min	max $\lvert\rho_\Delta\rvert$	max $\lvert\rho_L\rvert$
SNACKPACK	0.556	0.006	0.413	0.006	0.927	0.920
PEBBLES	0.402	0.002	0.513	0.000	0.509	0.767
UV_VISOR	0.208	0.195	0.620	0.013	0.017	0.893
MICROCHIP	0.207	0.196	0.677	0.014	0.016	0.927
TRANSLATOR	0.204	0.196	0.464	0.034	0.016	0.705
GALAXY_SOUNDS	0.204	0.196	0.425	0.041	0.020	0.648
OXYGEN_SHAKE	0.204	0.196	0.386	0.034	0.009	0.583
PANEL	0.204	0.196	0.536	0.025	0.013	0.750
SLEEP_POD	0.204	0.196	0.591	0.022	0.013	0.856
ROBOT	0.203	0.197	0.546	0.026	0.010	0.793

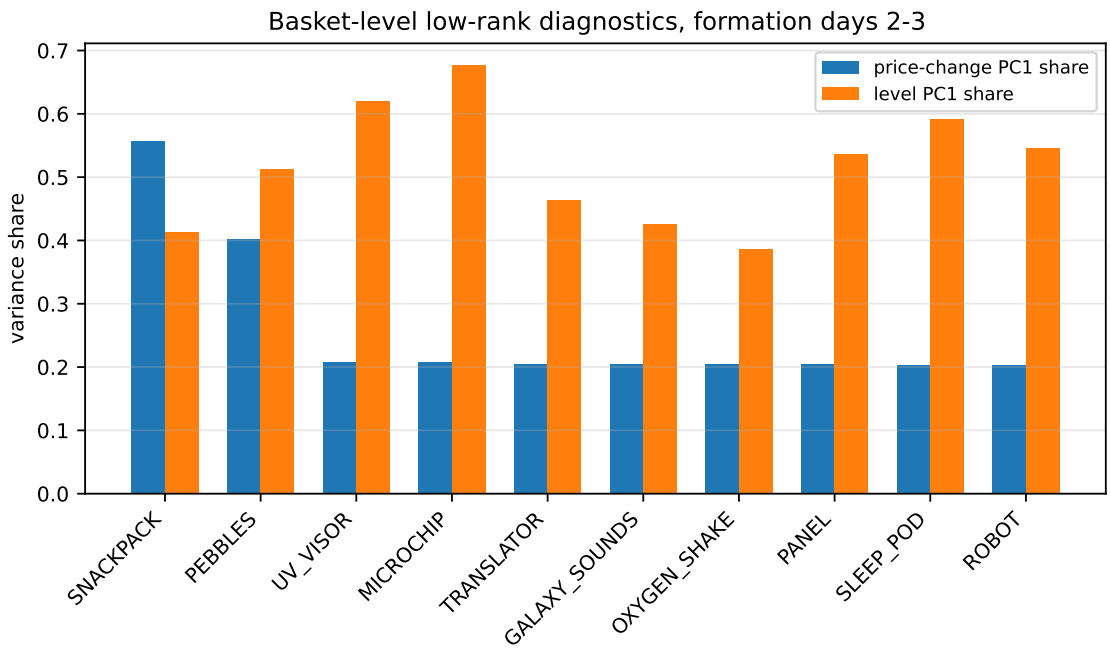

Figure 1. Basket-level variance shares. SNACKPACK has the clearest common-factor structure in price changes. Several secondary baskets have level common factors but almost independent price changes.

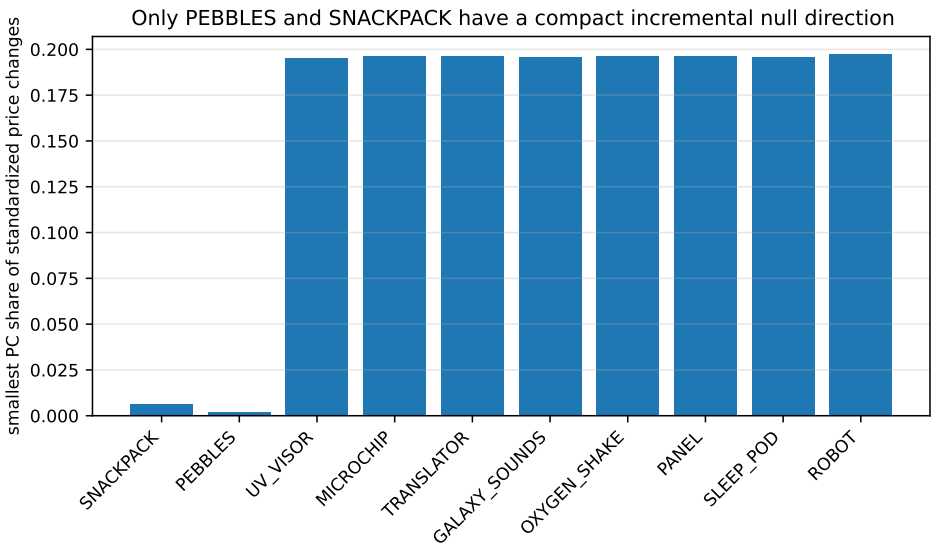

Figure 2. Smallest principal-component share of standardized price changes. This is the most compact visual proof that SNACKPACK and PEBBLES are structurally different from the other eight baskets under constant-incremental tests.

SNACKPACK's structure is not one spread; it is a relationship space. Figure 3 shows the formation price-change correlation matrix. CHOCOLATE and VANILLA are strongly negatively correlated. PISTACHIO and STRAWBERRY are strongly positively correlated, while RASPBERRY is strongly negatively correlated with both. This produces one CHOCOLATE–VANILLA spread and two independent spreads in the PISTACHIO–RASPBERRY–STRAWBERRY subgroup.

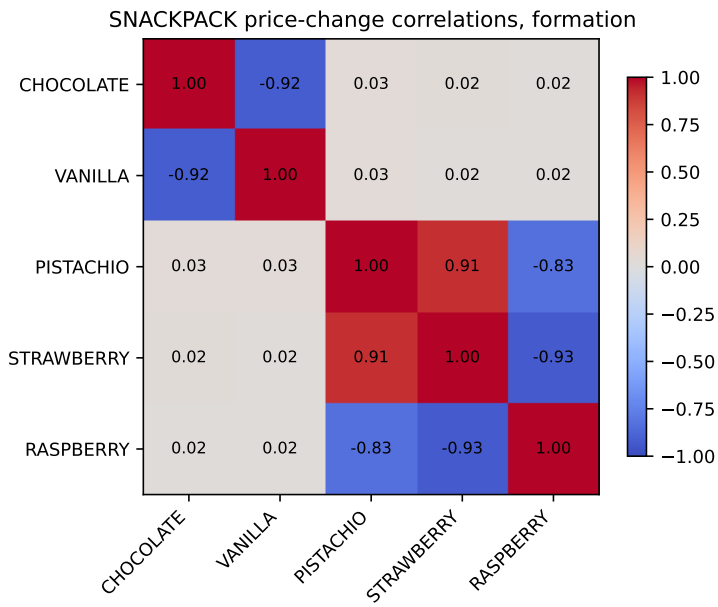

Figure 3. SNACKPACK price-change correlations in formation. The dominant structures are CHOCOLATE–VANILLA and the PISTACHIO–RASPBERRY–STRAWBERRY subgroup.

The secondary baskets fail the compact price-change test but have visible level structure. Figure 4 shows level correlations for MICROCHIP, SLEEP POD, OXYGEN SHAKE, TRANSLATOR, GALAXY SOUNDS, and UV VISOR. This matters because level co-movement can support a moving fair value even when first differences are weak. It also creates a false-positive risk: UV VISOR, PANEL, and ROBOT can be made to look calmer under a dynamic filter, but they do not have enough incremental structure to be promoted.

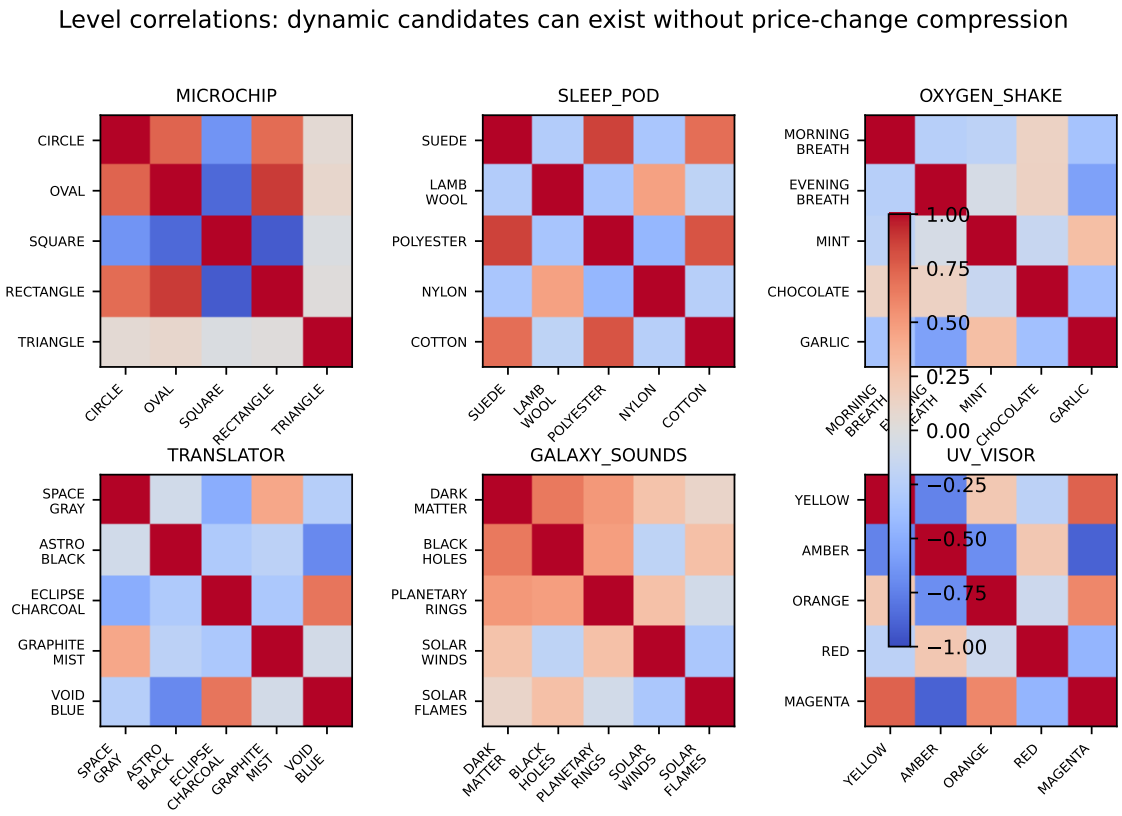

Figure 4. Level correlations for selected baskets. Level structure motivates dynamic fair values for MICROCHIP, SLEEP POD, OXYGEN SHAKE, and TRANSLATOR. Level correlation alone is not enough; UV VISOR is included as a counterexample.

PEBBLES is different from every other basket. It satisfies the additive identity

$$p_{XS,t} + p_{S,t} + p_{M,t} + p_{L,t} + p_{XL,t} \approx 50{,}000.$$

The identity residual has formation standard deviation 2.787, locked-OOS 95th absolute percentile 1.0, and locked-OOS maximum absolute deviation 18.5. The median half-spread of the five-leg taker trade is 32.5, which is 32.5 times the day-4 95th percentile of the identity error and 11.66 times the formation standard deviation. Therefore PEBBLES is a genuine fair-value identity but not a full-basket market-taking arbitrage.

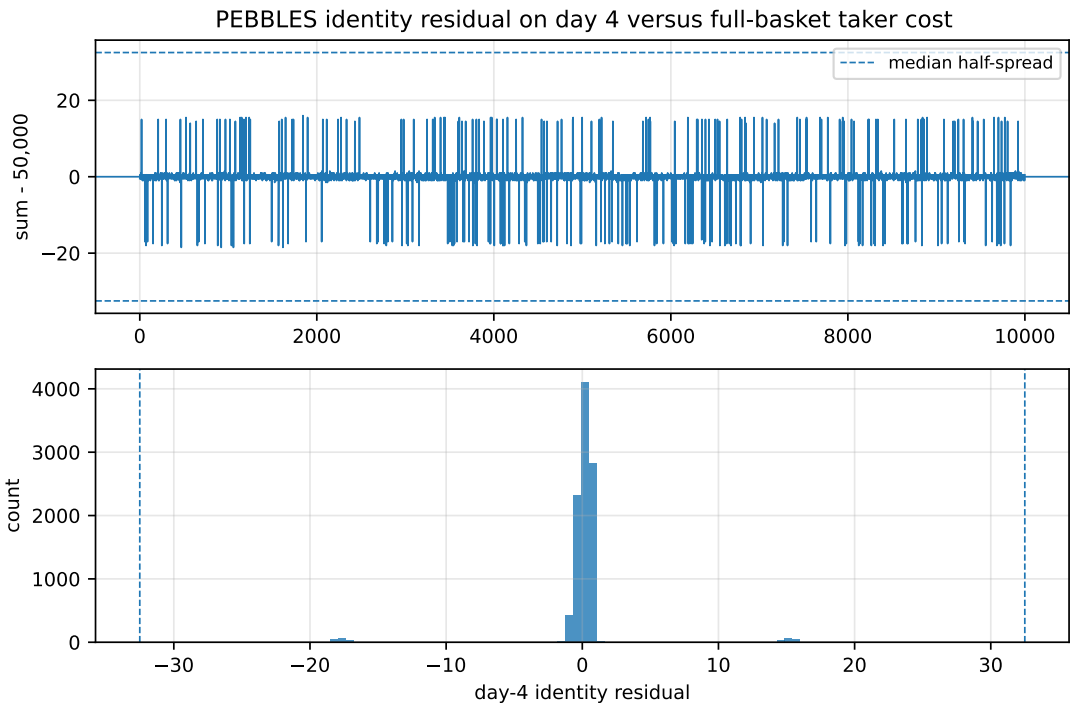

Figure 5. PEBBLES identity residual and full-basket taker half-spread. The identity is statistically exact enough to be used for fair value, but crossing all five legs is uneconomic.

3. Spread modelization and construction

Table 2 compares the best fixed or trend fair value against the causal dynamic fair value for the retained spreads. The strongest dynamic rescues are MICRO_LVL4, SNACK_CV, TRANS_SEG, and GALAXY_BHDM: their OOS median absolute z-scores fall by 91%, 65%, 71%, and 82%, respectively. GALAXY is still not promoted, because its trading behavior is unstable and its basket-level price-change evidence is weak. SLEEP_PC and SLEEP_SPC are less of a statistical rescue than an execution improvement: fixed or trend z-scores are not catastrophic, but the dynamic residual gives better trading behavior in the day-4 mid-price test.

Table 2. Fair-value model comparison on the same spread weights. "Fixed" is the better of static and deterministic trend. Gain is $1 - \text{med}|z|_{dyn}/\text{med}|z|_{fixed}$. Negative gain means the fixed/trend model had slightly better median calibration.

Spread	best fixed FV	fixed med $\|z\|$	fixed p95	dynamic med $\|z\|$	dynamic p95	gain
GALAXY_BHDM	trend	4.523	7.711	0.795	2.010	0.824
MICRO_LVL4	static	6.485	8.082	0.563	1.662	0.913
MICRO_OT	trend	0.733	1.657	0.395	1.125	0.461
OXY_ECG	trend	0.944	1.977	0.519	1.532	0.451
OXY_EG_POS	trend	1.677	2.767	0.699	2.078	0.583
OXY_MG	trend	0.417	1.450	0.470	1.587	-0.127
SLEEP_PC	static	0.603	1.318	0.584	1.970	0.031
SLEEP_SPC	trend	0.507	1.415	0.583	1.674	-0.150
SNACK_CV	trend	1.576	2.588	0.551	1.729	0.650
SNACK_META	trend	0.532	1.364	0.696	2.038	-0.308
SNACK_PS	static	1.790	2.304	1.131	2.209	0.368
SNACK_RS	static	1.509	2.010	0.880	1.936	0.417
TRANS_EV	trend	0.593	2.599	0.504	1.666	0.149
TRANS_SEG	trend	1.859	3.147	0.543	1.723	0.708

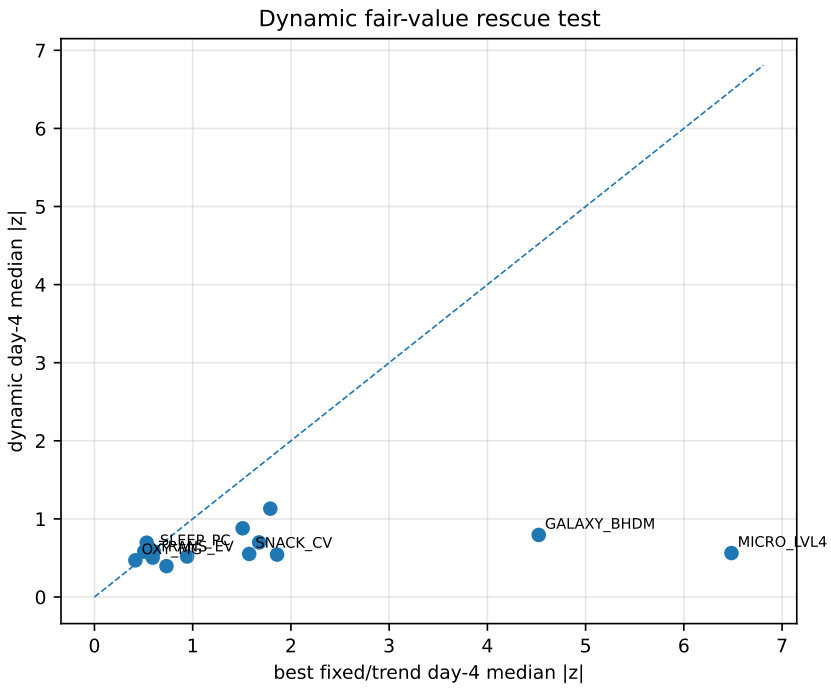

Figure 6. Static/trend versus dynamic day-4 residual stress. Points below the diagonal improve under causal dynamic fair value.

Table 3 gives the selected residual diagnostics. The most important columns are OOS median $|z|$, OOS 95th percentile $|z|$, $P(|z| > 2)$, and the formation ADF p-value. The table shows that most active residuals remain calibrated on day 4: MICRO_OT, OXY_MG, OXY_ECG, TRANS_EV, TRANS_SEG, SLEEP_PC, and SLEEP_SPC have OOS median $|z|$ below 0.60. SNACK_PS is slower and more persistent; it is kept because it is part of the SNACKPACK subgroup and because its threshold tests remain profitable. GALAXY_BHDM is deliberately left as watchlist despite a good dynamic z-score because it fails the stronger basket-level and trading-consistency bar.

Table 3. Final residual diagnostics. σ_F, ϕ_F, and half-life are estimated on formation. OOS columns are computed on locked day 4 with no refit.

Spread	Basket	FV	K	σ_F	ϕ_F	HL	OOS med $\|z\|$	OOS p95 $\|z\|$	$P(\|z\| > 2)$	ADF p
PEBBLES_ID	PEBBLES	identity		2.787	0.013	0.160	0.158	0.381	0.029	$< 10^{-4}$
SNACK_CV	SNACKPACK	dynamic	0.005	8.449	0.987	53.232	0.551	1.729	0.026	$< 10^{-4}$
SNACK_PS	SNACKPACK	dynamic	0.0002	49.652	0.999	1279.077	1.131	2.209	0.061	0.058
SNACK_RS	SNACKPACK	dynamic	0.002	17.285	0.996	165.380	0.880	1.936	0.041	0.000
SNACK_META	SNACKPACK	trend		29.902	0.999	861.121	0.532	1.364	0.001	0.032
MICRO_LVL4	MICROCHIP	dynamic	0.005	69.430	0.993	104.266	0.563	1.662	0.019	$< 10^{-4}$
MICRO_OT	MICROCHIP	dynamic	0.002	142.004	0.997	268.662	0.395	1.125	0.000	$< 10^{-4}$
SLEEP_PC	SLEEP_POD	dynamic	0.002	116.032	0.998	292.360	0.584	1.970	0.046	$< 10^{-4}$
SLEEP_SPC	SLEEP_POD	dynamic	0.002	107.718	0.998	350.444	0.583	1.674	0.014	0.000
OXY_MG	OXYGEN_SHAKE	dynamic	0.002	136.029	0.998	449.294	0.470	1.587	0.023	0.003
OXY_ECG	OXYGEN_SHAKE	dynamic	0.002	81.734	0.998	329.059	0.519	1.532	0.007	0.000
OXY_EG_POS	OXYGEN_SHAKE	dynamic	0.005	66.237	0.991	79.917	0.699	2.078	0.058	$< 10^{-4}$
TRANS_EV	TRANSLATOR	dynamic	0.002	106.887	0.998	298.749	0.504	1.666	0.014	$< 10^{-4}$
TRANS_SEG	TRANSLATOR	dynamic	0.001	78.220	0.998	333.247	0.543	1.723	0.013	0.000
GALAXY_BHDM	GALAXY_SOUNDS	dynamic	0.005	73.951	0.995	137.361	0.795	2.010	0.051	$< 10^{-4}$

Figure 7 plots the locked day-4 z-scores for the main retained spreads. The panels visually separate three cases. Core SNACKPACK spreads are slower but structured; secondary dynamic baskets have shorter, tradeable deviations; GALAXY is visibly less clean and therefore remains a watchlist signal.

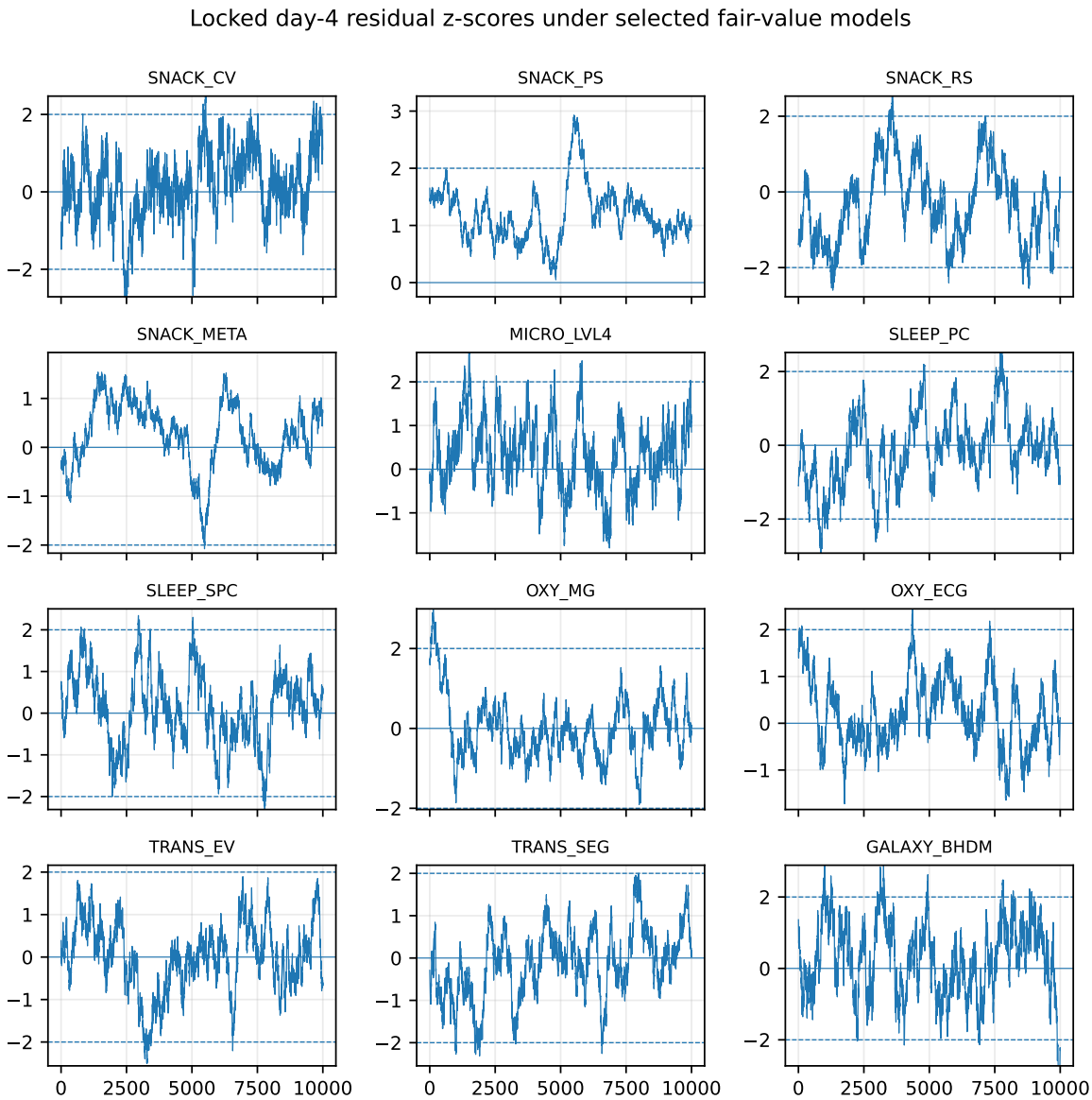

Figure 7. Locked day-4 z-scores for retained spreads under the selected fair-value model. Horizontal dashed lines mark ±2.

Table 4 addresses the false-positive issue. UV_AR, PANEL_24, and ROBOT_VI can produce apparently acceptable dynamic OOS z-scores. But their price-change null shares are 0.195–0.197, and their maximum absolute price-change correlations are only 0.010–0.017. Those numbers are consistent with independent innovations, not a structural intra-basket relation. They are not included in the final trade list.

Table 4. Rejected dynamic benchmarks. Good-looking dynamic z-scores are not enough when the basket has no price-change compression.

Spread	Basket	OOS med $\lvert z\rvert$	OOS p95 $\lvert z\rvert$	$P(\lvert z\rvert > 2)$	ADF p	Δ min	max $\lvert\rho_\Delta\rvert$	Level PC1
UV_AR	UV_VISOR	0.606	1.751	0.011	0.010	0.195	0.017	0.620
PANEL_24	PANEL	0.669	2.006	0.051	$< 10^{-4}$	0.196	0.013	0.536
ROBOT_VI	ROBOT	0.481	1.064	0.000	0.437	0.197	0.010	0.546

4. Trading construction

For a residual z_t, the sign convention is

$$z_t > z_{entry} : \text{ short the positive-weight legs and long the negative-weight legs,}$$

$$z_t < -z_{entry} : \text{ long the positive-weight legs and short the negative-weight legs.}$$

Positions exit when $|z_t| < z_{exit}$, with $z_{exit} = 0.5$ in the tests. The reported PnL is mid-price mark-to-market PnL in one L1-normalized spread unit; transaction costs are deliberately excluded except for the PEBBLES full-basket cost test. This is the correct comparison for deciding which residuals deserve later integration into a multi-strategy inventory book.

We compare three trading styles.

(1) *Hysteresis threshold*: enter at $|z| > z_{entry}$ and exit at $|z| < 0.5$.
(2) *Continuous linear*: use $q_t = -\operatorname{clip}(z_t/2, -1, 1)$ as a spread-unit target.
(3) *Passive fair-value quoting*: used only for PEBBLES, where the identity gives one-leg fair values instead of a full five-leg taker spread.

Table 5 compares the modelization choices at entry 2.0 for the main spreads. The table is not used to blindly select the maximum OOS PnL; it is used to detect whether a spread is stable across construction choices. MICRO_LVL4 is a clear example where static and trend fail and dynamic is required. SLEEP_PC and SLEEP_SPC are examples where dynamic is chosen for trading behavior even though fixed/trend residual medians are not disastrous. OXY_MG is a special case: its trend residual is statistically clean but does not fire at a 2.0 threshold; the spread is better used as a lower-threshold or continuous confirmation signal.

Table 5. Modelization and trading comparison at entry 2.0, exit 0.5. "medz" is day-4 median $|z|$; PnL is mid-price spread-unit PnL.

Spread	stat medz	stat PnL	trend medz	trend PnL	dyn medz	dyn PnL	Choice
SNACK_CV	1.81	48.78	1.58	-3.10	0.55	63.20	dynamic
SNACK_PS	1.79	-57.70	2.36	210.55	1.13	132.40	dynamic
SNACK_RS	1.51	13.15	3.46	140.42	0.88	71.41	dynamic
SNACK_META	1.33	100.62	0.53	47.00	0.70	4.75	trend
MICRO_LVL4	6.48	-1097.47	6.97	-1138.08	0.56	252.43	dynamic
MICRO_OT	1.64	146.75	0.73	75.00	0.40	0.00	dynamic
SLEEP_PC	0.60	0.00	2.11	531.00	0.58	697.25	dynamic
SLEEP_SPC	0.52	260.96	0.51	270.40	0.58	577.00	dynamic
OXY_MG	2.12	-508.75	0.42	0.00	0.47	-71.00	trend
OXY_ECG	1.44	94.50	0.94	204.12	0.52	232.12	dynamic
OXY_EG_POS	2.38	-359.98	1.68	-210.61	0.70	401.34	dynamic
TRANS_EV	1.75	-53.50	0.59	544.00	0.50	95.50	dynamic
TRANS_SEG	3.29	-238.38	1.86	132.50	0.54	483.00	dynamic
GALAXY_BHDM	4.96	-898.00	4.52	-824.00	0.79	-1.25	watch

Table 6 compares threshold entries for the selected active spreads. The strongest tactical secondary basket is SLEEP POD: SLEEP_PC earns 793.0 at entry 1.5 and 697.2 at entry 2.0, while SLEEP_SPC earns 573.7 and 577.0. TRANS_EV and TRANS_SEG both prefer the 1.5 threshold. MICRO_LVL4 prefers the stricter 2.0 threshold. OXY_MG should not be traded as a high-threshold signal; OXY_ECG and OXY_EG_POS are better active OXY expressions.

Table 6. Threshold comparison on locked day 4. Entries use exit 0.5 and one spread unit.

Spread	PnL 1.5	PnL 2.0	PnL 2.5
MICRO_LVL4	159.3	252.4	62.4
OXY_ECG	235.8	232.1	0.0
OXY_MG	253.0	-71.0	23.5
SLEEP_PC	793.0	697.2	386.2
SLEEP_SPC	573.7	577.0	0.0
SNACK_META	91.9	47.0	0.0
SNACK_PS	310.7	132.4	156.7
TRANS_EV	446.0	95.5	71.0

Table 7 gives the recommended trading settings. The linear PnL column is included because some residuals, especially MICRO_OT and OXY_MG, are better as continuous sizing/confirmation signals than as sparse high-threshold entry signals.

Table 7. Recommended trading settings. Mid PnL ignores costs except that PEBBLES is excluded from this table as a full-basket taker trade.

Spread	entry	entries	mid PnL	max DD	exposure	linear PnL
SNACK_CV	2.00	10	63.20	21.68	0.106	408.11
SNACK_PS	1.50	4	310.71	67.97	0.601	199.53
SNACK_RS	2.00	6	71.41	50.53	0.241	224.08
SNACK_META	1.50	3	91.88	21.88	0.137	100.23
MICRO_LVL4	2.00	7	252.43	190.27	0.095	212.23
MICRO_OT	2.00	0	0.00	0.00	0.000	720.93
SLEEP_PC	2.00	7	697.25	255.25	0.155	643.68
SLEEP_SPC	2.00	6	577.00	160.06	0.148	524.70
OXY_MG	1.50	5	253.00	286.25	0.132	456.04
OXY_ECG	2.00	3	232.12	97.50	0.078	588.44
OXY_EG_POS	2.00	14	401.34	260.49	0.197	707.65
TRANS_EV	1.50	9	446.00	303.75	0.206	566.26
TRANS_SEG	1.50	8	450.25	129.00	0.171	549.44
GALAXY_BHDM	1.50	32	531.50	362.25	0.356	52.85

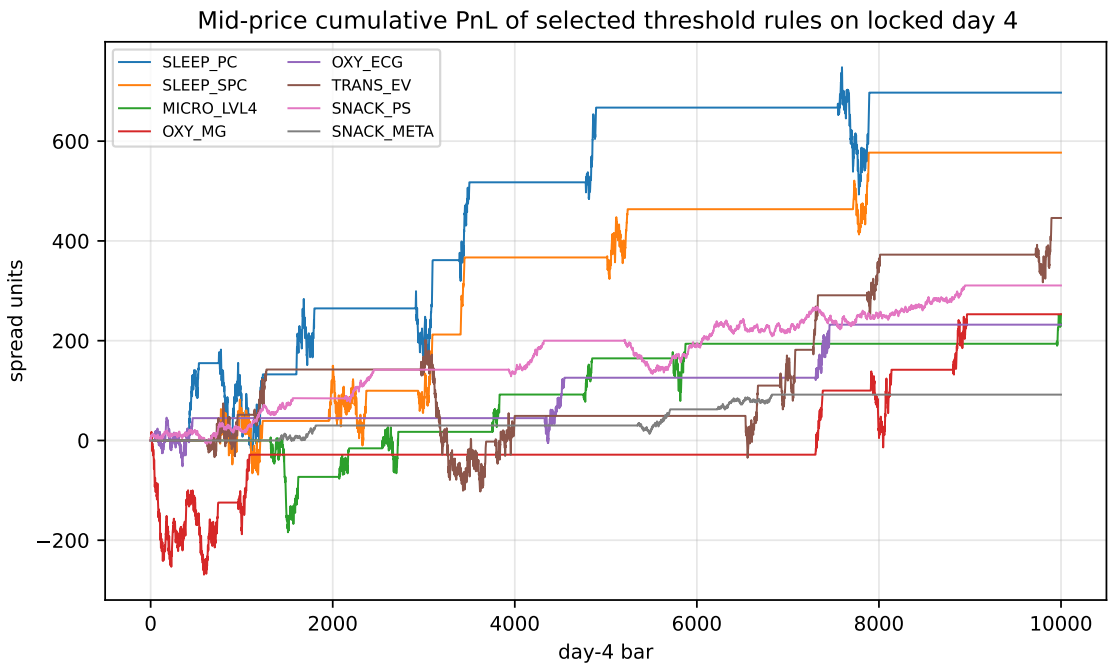

Figure 8. Cumulative mid-price PnL for selected threshold rules on locked day 4. These curves are spread-unit PnL and are not cost adjusted.

5. Final spread set

Table 8 is the finalized intra-basket list. PEBBLES is not a full-basket market-taking strategy; it is a fair-value quoting identity. SNACKPACK contains the core relationship space. SLEEP POD, MICROCHIP, OXYGEN SHAKE, and TRANSLATOR are secondary dynamic strategies. GALAXY remains a watchlist. UV VISOR, PANEL, and ROBOT are excluded.

Table 8. Final spread definitions and model choices. Non-PEBBLES spreads are L1-normalized internally before scoring and trading.

Spread	Formula	FV model	Verdict
PEBBLES_ID	XS+S+M+L+XL-50000	identity	Quote only FV
SNACK_CV	CHOCOLATE+0.928 VANILLA	dynamic, K=0.005	Trade
SNACK_PS	PISTACHIO-0.590 STRAWBERRY	dynamic, K=0.0002	Trade
SNACK_RS	RASPBERRY+0.922 STRAWBERRY	dynamic, K=0.002	Trade
SNACK_META	CHOCOLATE+VANILLA-PISTACHIO-RASPBERRY	trend	Trade
MICRO_LVL4	0.23 CIRCLE-OVAL+0.90 RECTANGLE+0.21 TRIANGLE	dynamic, K=0.005	Trade
MICRO_OT	OVAL-TRIANGLE	dynamic, K=0.002	Confirmation
SLEEP_PC	POLYESTER-COTTON	dynamic, K=0.002	Trade
SLEEP_SPC	0.61 SUEDE-POLYESTER+0.54 COTTON	dynamic, K=0.002	Trade
OXY_MG	MORNING_BREATH-GARLIC	dynamic, K=0.002	Trade
OXY_ECG	MORNING-EVENING+CHOCOLATE-GARLIC	dynamic, K=0.002	Trade
OXY_EG_POS	0.71 EVENING_BREATH+0.29 GARLIC	dynamic, K=0.005	Overlay/watch
TRANS_EV	ECLIPSE_CHARCOAL-VOID_BLUE	dynamic, K=0.002	Trade
TRANS_SEG	SPACE+ECLIPSE-GRAPHITE-VOID	dynamic, K=0.001	Trade
GALAXY_BHDM	BLACK_HOLES-DARK_MATTER	dynamic, K=0.005	Watchlist

The implementation logic for each class is as follows.

PEBBLES. Do not cross all five legs. Use

$$\mathrm{FV}_i(t) = 50{,}000 - \sum_{j \neq i} p_{j,t}$$

for one-leg quoting. Buy product i only when $\mathrm{FV}_i(t) - a_{i,t}$ is large enough; sell product i only when $b_{i,t} - \mathrm{FV}_i(t)$ is large enough. The spread identity is real, but the median full-basket taker half-spread is 32.5 versus a day-4 95th absolute identity error of 1.0.

SNACKPACK. Trade the relationship space, not one single spread. The active spreads are SNACK_CV, SNACK_PS, SNACK_RS, and SNACK_META. Because these residuals overlap in components, final inventory should be computed by summing spread targets,

$$q_t^{prod} = \sum_j q_{j,t} w_j,$$

then applying product-level position controls. SNACK_CV should use the dynamic residual; SNACK_META should use the formation trend residual.

SLEEP POD. Promote SLEEP_PC and SLEEP_SPC as active secondary strategies. Both should use the causal dynamic residual with $K = 0.002$. Entry 2.0 is a good default; entry 1.5 increases signal count and still works for SLEEP_PC.

MICROCHIP. MICRO_LVL4 is the active trading spread and should use dynamic fair value with $K = 0.005$ and entry 2.0. MICRO_OT has excellent residual calibration but few high-threshold entries; use it as confirmation or continuous sizing rather than as an independent sparse trigger.

OXYGEN SHAKE. The earlier CHOCOLATE-centered view is incomplete. The statistical screen finds a cleaner signed link involving MORNING_BREATH and GARLIC and a stronger four-leg rotation MORNING–EVENING+CHOCOLATE–GARLIC. OXY_MG is best as a lower-threshold or linear confirmation signal; OXY_ECG is the cleaner active signed rotation. OXY_EG_POS has good dynamic behavior and PnL, but it is an all-positive sub-basket average, so it should be used only when directional inventory in the two positive legs is acceptable or offset elsewhere.

TRANSLATOR. TRANS_EV and TRANS_SEG are the preferred dynamic spreads. TRANS_EV is the simpler tactical pair; TRANS_SEG is the stronger four-leg ladder rotation. Both prefer entry 1.5 in the day-4 test.

GALAXY SOUNDS. GALAXY_BHDM is a watchlist signal only. Dynamic fair value reduces residual stress, but fixed/trend models fail badly, the basket does not have price-change compression, and the linear sizing test is weak. It should not be promoted until it is confirmed by a cross-basket or lead-lag signal.

Conclusion. The final hierarchy is: PEBBLES for passive identity quoting; SNACKPACK as the core multi-spread basket; SLEEP POD, MICROCHIP, OXYGEN SHAKE, and TRANSLATOR as secondary dynamic-fair-value spreads; GALAXY as watchlist; and UV VISOR, PANEL, ROBOT as rejected. The main modeling lesson is that w and m_t are different objects. The spread weights describe the basket relationship; the fair-value process describes where that relationship should be priced now. Treating every spread as a fixed-mean pair would miss the dynamic opportunities, while letting every basket use a dynamic filter would create false positives.